

Assignment 3: Data Exploration

Tracy Che

OVERVIEW

This exercise accompanies the lessons in Environmental Data Exploration M3: Data Exploration.

Directions

1. Rename this file `<FirstLast>_A03_DataExploration.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Assign a useful **name to each code chunk** and include ample **comments** with your code.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.
7. After Knitting, submit the completed exercise (PDF file) to the dropbox in Canvas.

TIP: If your code extends past the page when knit, tidy your code by manually inserting line breaks.

TIP: If your code fails to knit, check that no `install.packages()` or `View()` commands exist in your code.

Set up your R session

1. Load necessary packages (tidyverse, lubridate, here), check your current working directory and upload two datasets: the ECOTOX neonicotinoid dataset (ECOTOX_Neonicotinoids_Insects_raw.csv) and the Niwot Ridge NEON dataset for litter and woody debris (NEON_NIWO_Litter_massdata_2016-26_raw.csv). Name these datasets “Neonics” and “Litter”, respectively.

```
# Load the packages required for this assignment
library(tidyverse)
library(lubridate)
library(here)
```

```
# Check the working directory and project root
getwd()
```

```
## [1] "/home/guest/tracyche"
```

```
here()
```

```
## [1] "/home/guest/tracyche"
```

```
# Import the ECOTOX dataset
Neonics <- read.delim(
  here("Data", "ECOTOX_Neonicotinoids_Insects_raw.txt"),
  sep = "|",
  header = TRUE,
  stringsAsFactors = FALSE,
  check.names = FALSE
)

# Remove extra spaces from column names and make them valid R names
names(Neonics) <- make.names(
  trimws(names(Neonics)),
  unique = TRUE
)

# Import the NEON dataset
Litter <- read.csv(
  here("Data", "NEON_NIWO_Litter_massdata_2016-26_raw.csv"),
  stringsAsFactors = FALSE,
  na.strings = c("", "NA")
)
```

```
## Learn about your system
```

2. The neonicotinoid dataset was collected from the Environmental Protection Agency's ECOTOX Knowledgebase.

> Answer: Neonicotinoids are used to control insect pests, but exposure can also affect beneficial, non-target organisms.

3. The Niwot Ridge litter and woody debris dataset was collected from the National Ecological Observatory Network.

> Answer: Litter and woody debris provide information about material moving from vegetation to the forest floor.

4. How is litter and woody debris sampled as part of the NEON network? Read the NEON_Litterfall_UserGuide.

> Answer:

1. Trap design
2. Sampling frequency
3. Sampling processing

```
## Obtain basic summaries of your data (Neonics)
```

5. What are the dimensions of the dataset?

```
''' r
# Report the number of rows and columns
dim(Neonics)
```

```
## [1] 8748 76
```

```
nrow(Neonics)
```

```
## [1] 8748
```

```
ncol(Neonics)
```

```
## [1] 76
```

6. Using the `summary` function on the “Effect” column, determine the most common effects that are studied. Why might these effects specifically be of interest? [Tip: The `sort()` command is useful for listing the values in order of magnitude...]

```
# Count records in each effect category
effect_counts <- summary(
  factor(Neonics$Effect),
  maxsum = Inf
)

# Sort the counts from largest to smallest
effect_counts <- sort(
  effect_counts,
  decreasing = TRUE
)

# Display the six most frequent effects
head(effect_counts, 6)
```

```
##      Mortality      Population      Genetics      Behavior
##           2687           1377           905           826
## Feeding behavior      Reproduction
##           712           557
```

Answer: The most common effects that are studied is morality, population, genetics, behavior, feeding behavior, and reproduction. These effects are more direct and can infer a lot easy-understanding aspects of insects.

7. Using the `summary` function, determine the six most commonly studied species in the dataset (common name). What do these species have in common, and why might they be of interest over other insects? Feel free to do a brief internet search for more information if needed.[TIP: Explore the help on the `summary()` function, in particular the `maxsum` argument...]

```
# Count records for every common-name category
species_counts <- summary(
  factor(Neonics$Species.Common.Name),
  maxsum = Inf
)

# Rank common names by their number of records
species_counts <- sort(
  species_counts,
  decreasing = TRUE
)
```

```
)

# Display the six most frequent common names
head(species_counts, 6)
```

```
##           Honey Bee Carniolan Honey Bee           Parasitic Wasp           Bumble Bee
##           2310                622                428                360
##   Italian Honeybee   Asiatic Honey Bee
##           348                345
```

Answer: They are honey bee, carniolan honey bee, parasitic wasp, bumble bee, Italian honeybee, and Asiatic honeybee. Bees species are most commonly studied. Bees are more related to human life, such as producing honey, and they are also artificially raised a lot, so this could be the reason they are more studied than other species.

8. Concentrations are always a numeric value. What is the class of `Conc.1.Mean..Author..` column in the dataset, and why is it not numeric? [Tip: Viewing the dataframe may be helpful...]

```
# Check the class of the column
class(Neonics$Conc.1.Mean..Author.)
```

```
## [1] "character"
```

```
# Inspect entries in the original data
head(unique(Neonics$Conc.1.Mean..Author.), 12)
```

```
## [1] "0.025/" "144.0/" "0.267" "0.295" "20.636" "76.035" "12/"
## [8] "9.6"    "0.25/" "197.625" "39.5/" "3.9"
```

Answer: The data includes `'/'`, which means a comment in ECOTOX. Therefore it cannot be purely numeric.

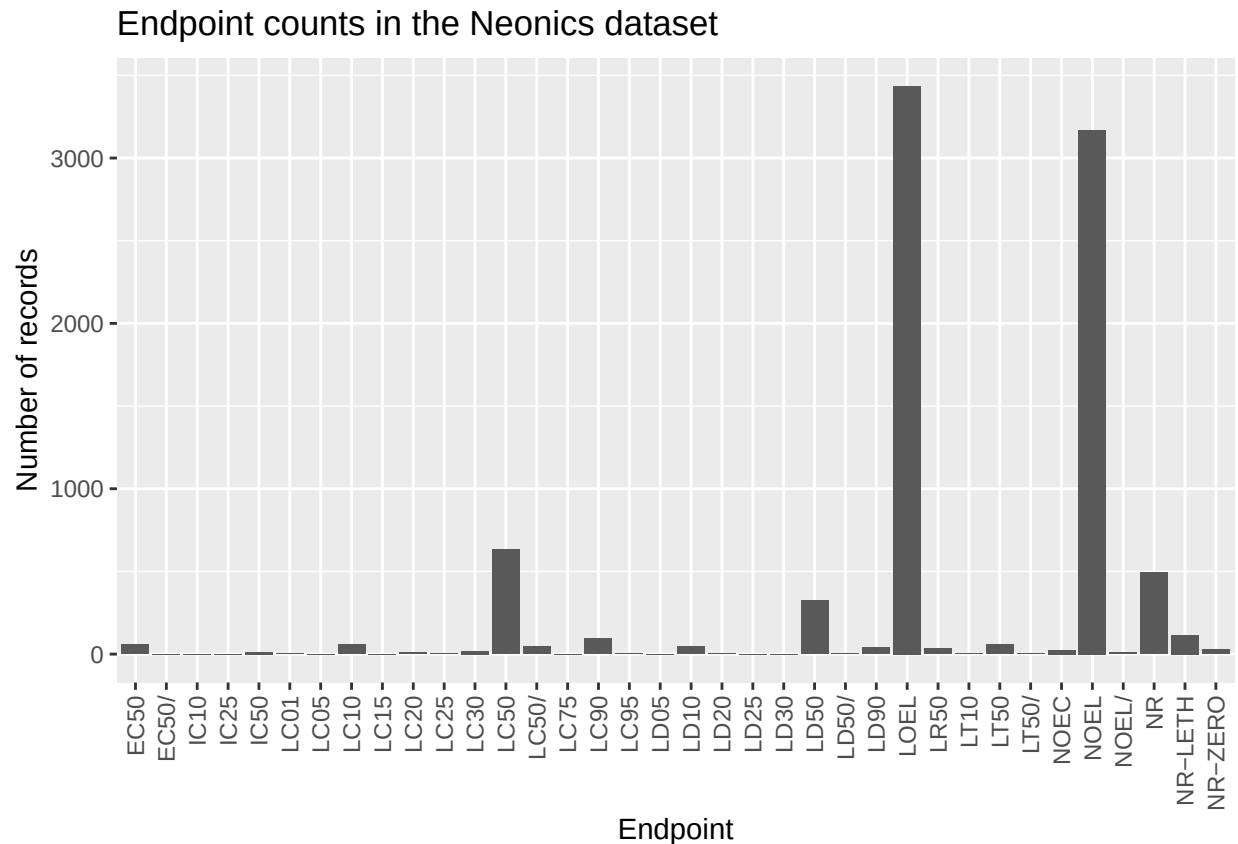
Explore your data graphically (Neonics)

9. Create a bar graph of Endpoint counts. What are the two most common end points, and how are they defined? Consult the ECOTOX_CodeAppendix for more information.

[TIP: Add `theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))` to the end of your plot command to rotate and align the X-axis labels...]

```
# Plot the number of records for each endpoint
ggplot(Neonics, aes(x = Endpoint)) +
  geom_bar() +
  labs(
    title = "Endpoint counts in the Neonics dataset",
    x = "Endpoint",
    y = "Number of records"
  ) +
  theme(
```

```
axis.text.x = element_text(
  angle = 90,
  vjust = 0.5,
  hjust = 1
)
)
```



Answer: The two most frequent end points are LOEL and NOEL. LOEL represents the Lowest-Observable-Effect-Level. NOEL means No-Observable-Effect-Level.

Explore your data (Litter)

- Determine the class of collectDate. Is it a date? If not, change to a date and confirm the new class of the variable. Using the `unique` function, determine which dates litter was sampled in August 2018.

```
# Check the original class
class(Litter$collectDate)
```

```
## [1] "character"
```

```
# Convert text to Date objects
Litter$collectDate <- ymd(Litter$collectDate)
```

```
# Confirm the new class
class(Litter$collectDate)
```

```
## [1] "Date"
```

```
# Check that the conversion did not produce missing dates  
sum(is.na(Litter$collectDate))
```

```
## [1] 0
```

```
# Select records collected in August 2018  
august_2018 <- Litter %>%  
  filter(  
    year(collectDate) == 2018,  
    month(collectDate) == 8  
  )
```

```
# List the distinct sampling dates  
sort(unique(august_2018$collectDate))
```

```
## [1] "2018-08-02" "2018-08-30"
```

11. Using the `unique` function, list the different plotIDs sampled at Niwot Ridge. How is the information obtained from `unique` different from that obtained from `summary`?

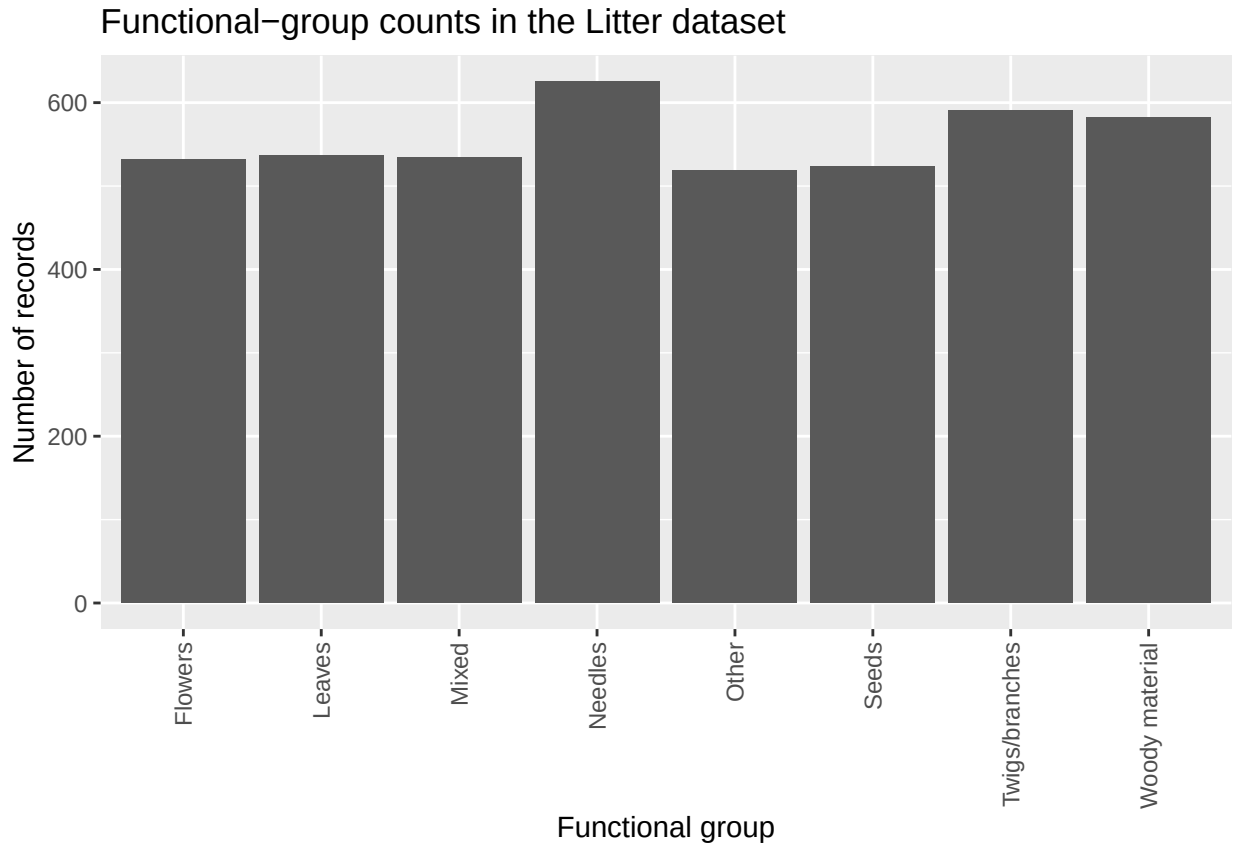
```
# List each plot identifier once  
sort(unique(Litter$plotID))
```

```
## [1] "NIWO_040" "NIWO_041" "NIWO_046" "NIWO_047" "NIWO_051" "NIWO_057"  
## [7] "NIWO_058" "NIWO_061" "NIWO_062" "NIWO_063" "NIWO_064" "NIWO_067"
```

Answer: `unique` function lists the distinct plotIDs without repeating them. In contrast, `summary` function reports the number of records associated with each plot. Therefore, ‘`unique`’ identifies the plots, while the factor ‘`summary`’ also describes how frequently each plot appears in the data.

12. Create a bar graph of functionalGroup counts. This shows you what type of litter is collected at the Niwot Ridge sites. Notice that litter types are fairly equally distributed across the Niwot Ridge sites.

```
# Plot the number of records in each functional group  
ggplot(Litter, aes(x = functionalGroup)) +  
  geom_bar() +  
  labs(  
    title = "Functional-group counts in the Litter dataset",  
    x = "Functional group",  
    y = "Number of records"  
  ) +  
  theme(  
    axis.text.x = element_text(  
      angle = 90,  
      vjust = 0.5,  
      hjust = 1  
    )  
  )  
)
```



What type(s) of litter tend to have the highest biomass at these sites?

Answer: Needles have highest biomass, and twigs/branches and woody material also have high biomass at these sites.

13. Write a short data-exploration summary for Neonics.

- **One key characteristic:** The dataset contains 8,748 result records from a selected terrestrial Insects/Spiders extract, covering multiple neonicotinoids, organisms, effects, and test conditions.
- **One data-quality issue:** The field contains comment markers, so it is stored as text rather than numeric values. Needs to be treated carefully before numerical analysis.
- **One interesting pattern:** Different types of bees are very popular among the studies, such as honey bees and bumble bees.
- **One question for further analysis:** Should the different bee types be treated similarly and specifically than other insect species? > Why is the study frequency of bees much higher than other insects?

14. Write a short data-exploration summary for Litter.

- **One key characteristic:** The dataset contains 4,443 mass records from 12 NIWO plots and eight material categories, with collection dates spanning June 2016 to October 2024.

- **One data-quality issue:** The file includes 354 QA reweighing records. These are not necessarily errors, but treating them as independent samples would duplicate some measurements in the analysis.
- **One interesting pattern:** The biomass of each content groups are quite even across the area, showing that litter types are varied but still balanced.
- **One question for further analysis:** Is there a potential reason that needles are more recorded than other types?